Cheatsheet

```
# Build and tag an image
docker build -t <image-name> <build-context>

# Start a container using its image name
docker run <image-name>

# Start a container in background
docker run -d <image-name>

# Display all running containers
docker ps

# Stop a container
docker stop <container-id>

# Access a running container
docker exec -it <container-id> /bin/sh

# Create a Docker network
docker network create <name>

# Start a container and override the entry point
docker run --entrypoint /bin/sh <image-name>

# Start a container and override the command
docker run <image-name> <command>

docker run -d --rm -p --network <networkName> 8080:8080 plantuml/plantuml-server
-d = run in background
--rm = remove the container on stop
-p HOST:CONTAINER = map machin port 8080 to this container port 8080

# Delete all stopped containers
docker container prune

# Delete all images
docker image prune

# run compose
docker compose up -d
-d = run in background

# stop and remove compose
docker compose down

#IF YOU WANT TO exec, run, ps, ... -> JUST ADD 'compose' before the command
```

Cloud computing

Not to be confused with Cloud Computing (horse).

Create a docker file

```
# Image sur la quelle on ce base pour crée la notre
FROM ubuntu:24.04
# Pour une app java FROM eclipse-temurin:21-jre

# Crée une variable d'environement
ENV key=value

# copie un directory (Sur notre machine) dans un autre (dans l'image)
COPY sourceDir destDir

# en gros, c'est un cd dans l'imagine
WORKDIR path

# Exec command
CMD ["echo", "Hello, World!"]

# Expose port
EXPOSE 8000
```

Docker compose

OK, en gros, un docker compose c'est un fichier qui sert à crée une infra, genre le container c'est les machines et le compose il dit just qu'il faut pour que ça marche et dans quel ordre les lancers (peut aussi servire à crée les images)

- `service` : Nom du service
- `image` : image à utiliser
- `port` : port a exposer (HOST:CONTAINER)
- `volumes` : mount volumes int the container
- `environment` : ENV variable
- `network` : network entre container

SSH key generation

Fait avec `ssh-keygen` (command) => crée une clef public et une clef privée
(public key = `<key>.pub` | private key = `<key>`)
Une passpharse peut être utiliser pour encrypter la private key

SCP

c'est un protocol utiliser pour copier des fichier d'une machine à une autre en remote (remplace FTP)

Example

```
networks:
  pantoufle:

services:
  ncat-server:
    hostname: my-server
    image: ncat
    command:
      - -l
      - "1234"
    networks:
      - pantoufle

  ncat-client:
    image: ncat
    command:
      - my-server
      - "1234"
    networks:
      - pantoufle
```

SSH & SCP

(SSH, GPC, etc.) => public/private keys security

SSH

SSH is a protocol use in TCP (port 22)
Uliser pour : remote machine connection, copy files (with SCP)

SSH is also the name of the software that implements the SSH protocol on most machine

SSH key algo

- RSA.
- DSA.
- **ECDSA**
- **Ed25519**

Most recent/secure

SSH key fingerprint

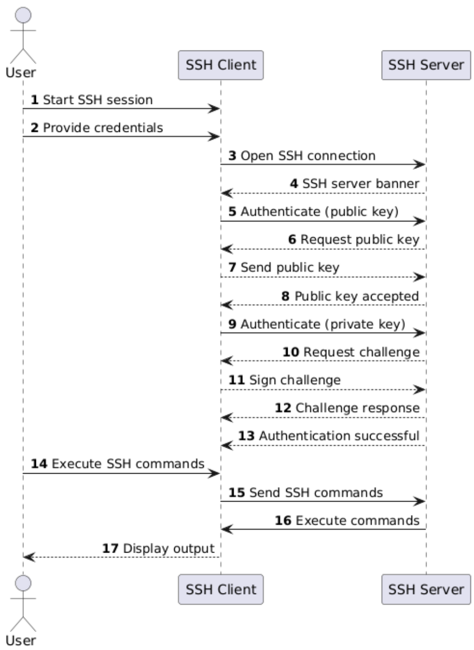
C'est le hahs de la public key (utiliser pour identifier la clef public)

ça permette de verifier que la clef utiliser est bien la bonne et d'eviter les **middle man attacks**

(Known hosts (hosts that you have allowed to connect to) are stored in the `~/.ssh/known_hosts` file)

1er connection : demande de confirmation du fingerprint de la clef public distant pour eviter le middle man attaque

SSH authentication process



HTTP & curl

Javalin (basé sur **Jetty**) (**Quarkus** ou **Spring Boot** pour la production)
curl = command line tool pour envoyer des requet (HTTP(S), FTP(S), SFTP, etc.)

```
curl -X POST -d "Hello, world!" http://localhost:8080/body
```

GET by default
-X = use HTTP
-v = print request headers
-i = print response header
-d = set http body (to "Hello, world!")
-H = set header | -H "Accept: text/plain"

Protocol

- Read-eval-print loop (REPL) (jsp où le mettre)

Define

- Aperçus : Résumer du problème que dois resoudre le protocol
- Protocol utiliser : UDP / TCP ?
- Message : lists de messages du protocol
- Command : List d'action possible
- Example : Example de routine en UML

HTTP

Proto **Host** **Query params**

https://community.steam.com/rivals/?id=350

Port (80 = HTTP | 443 = HTTPS) **Ressource Path**

`/users/{user-id}/view` | path parameter | parametrer intégré dans le chemin
`gaps.heig-vd.ch` -> `gaps` = subdomain | `heig-vd` = domain | `ch` = top-level domain

HTTP does not transfer objects, it transfers representations of objects. This means that the server can send the same resource in different representations.
The same resource, i.e. the URL /my-resource, can be sent in different representations:

Javalin

```java
public static void main(String[] args) {
    Javalin app = Javalin.create();
    // ### Path parameter ###
    app.get(
        "/content/{contentId}",
        ctx -> {
          String contentID = ctx.pathParam("contentId");
          ctx.result("GET "+contentID);
        });
    // ### query parameter ###
    app.get(
        "/content/{contentId}",
        ctx -> {
            //?id=test&data=prout
            String id = ctx.queryParam("id");
            String data = ctx.queryParam("data");
            if (id == null || data == null) {
                //io.javalin.http.BadRequestResponse;
                throw new BadRequestResponse();
            }
          ctx.result("GET "+id+" "+data);
        });
    app.post(
        "/data",
        ctx -> {
            // HTTP body
            String data = ctx.body();
            ctx.result("POST msg").status(HttpStatus.CREATED)
        });
    app.patch(
        "/",
        ctx ->
            ctx.result("PACTH msg").status(HttpStatus.OK));
    app.delete(
        "/",
        ctx ->
            ctx.result("DEL msg").status(HttpStatus.NO_CONTENT));

    app.start(8080);
}
```

WITH Content type

```java
app.get("/content/{contentId}",
    ctx -> {
        String acceptHeader = ctx.header("Accept");
        if (acceptHeader == null) {
            throw new BadRequestResponse();
        }

        if (acceptHeader.contains("text/html")) {
            ctx.contentType("text/html");
            ctx.result("<h1>Hello, world!</h1>");
        } else if (acceptHeader.contains("text/plain")) {
            ctx.contentType("text/plain");
            ctx.result("Hello, world!");
        } else {
            // io.javalin.http.NotAcceptableResponse;
            throw new NotAcceptableResponse();
        }
    });
```

COOKIE

```java
app.get("/content/{contentId}",
    ctx -> {
        // get cookie "cookie"
        String cookie = ctx.cookie("cookie");
        if (cookie == null) {
            // set cookie = "new-cookie"
            ctx.cookie("cookie", "new-cookie");
            ctx.result("COOKIE !!");
        } else {
            ctx.result("COOKIE value : '" + cookie + "'!");
        }
    });
```

Body Validator (use to manage missing data in the body)

```java
app.get("/sing-in/",
    ctx -> {

        Interger id = ctx.pathParamAsClass("id", Integer.class).get();

        User newUser = ctx.bodyValidator(User.class)
            .check(obj -> obj.email() != null, "Missing email")
            .get();

        users.put(newUser.id(), newUser);
        ctx.status(HttpStatus.CREATED);
        ctx.json(newUser);
    });
```

HTTP headers

Header	EX	Description
`Accept`	`text/html`	The media types accepted by the client
`Content-Type`	`text/html`	The media type of the body sent by the client or from the server
`Content-Length`	The length of the body sent by the client or the server	
`User-Agent`	The user agent of the client (the browser name and version, the curl version, etc.)	
`Host`	The host of the server	
`Set-Cookie`	The cookies set by the server	

URL encoding

`Super test`

↓

`Super%20test`

Request

- The HTTP method.
- The URL of the resource.
- The supported HTTP version.
- Some HTTP headers.
- The HTTP body (optional).
- The query parameters (optional).
- The cookies (optional).
- The content type (optional).

```
<HTTP method> <URL> HTTP/<HTTP version>
<HTTP headers>
<Empty line>
<HTTP body (optional)>
```

```
GET / HTTP/1
Host: gaps.heig-vd.ch
User-Agent: curl/8.1.2
Accept: */*
```

Host - The host of the server (gaps.heig-vd.ch in this case).
User-Agent - The user agent that sent the request (curl in this case).
Accept - The content types accepted by the user agent (any type in this case).

Respond

- The HTTP version.
- The HTTP status code.
- The HTTP headers.
- The HTTP body (optional).
- The cookies (optional).
- The content type (optional).
- The content length (in HTTP/1.1).
- The content encoding (optional).
- **Content-Length** - The length of the content in bytes (6111 bytes in this case).
- **Content-Type** - The content type of the resource (text/html in this case).

```
HTTP/<HTTP version> <HTTP status code> <HTTP status message>
<HTTP headers>
<Empty line>
<HTTP body>
```

HTTP Response statue code

- `1xx` - **Informational responses**
  - `101` - Switching Protocols (the server switches to a different protocol).
  - `102` - Processing (the server is processing the request).
- `2xx` - **Successful responses**
  - `200` - **OK** (the request was successful).
  - `201` - **Created** (new resource was created).
  - `202` - **Accepted** (the request was accepted but not yet processed).
  - `204` - **No Content** (the request was successful but the server does not send any content).
- `3xx` - **Redirection messages**
  - `301` - **Moved Permanently** (the resource has been moved permanently to a new URL).
  - `302` - **Found** (the resource has been moved temporarily to a new URL).
  - `304` - **Not Modified** (the resource has not been modified since the last request).
- `4xx` - **Client error responses**
  - `400` - **Bad Request** (the request is malformed).
  - `401` - **Unauthorized** (the request requires authentication).
  - `403` - **Forbidden** (the request is forbidden).
  - `404` - **Not Found** (the resource does not exist).
  - `405` - **Method Not Allowed** (the HTTP method is not allowed for this resource).
  - `409` - **Conflict** (the request could not be processed because of a conflict).
  - `410` - **Gone** (the resource is no longer available and has been removed).
  - `429` - **Too Many Requests** (the client has sent too many requests in a given amount of time).
- `5xx` - **Server error responses**
  - `500` - **Internal Server Error** (the server encountered an unexpected condition that prevented it from fulfilling the request).
  - `501` - **Not Implemented** (the server does not support the functionality required to fulfill the request)
  - `502` - **Bad Gateway** (the server received an invalid response from an upstream server).
  - `503` - **Service Unavailable** (the server is currently unable to handle the request due to a temporary overload or scheduled maintenance).
  - `504` - **Gateway Timeout** (the server did not receive a timely response from an upstream server).

  The default status code is `200` (OK). This means that you can omit the status code.

Proxy

Proxy = composent d'une infra qui est intermediaire au requets de resources

Forward | Client --> Server | Extern --> Interne

En gros, c'est composant infra qui vas redistribuer les request d'une group de client ver **n'import quel** servers
Configured to handle request from a group of clients to any other server.
Often used to corporate environments (security)

Reverse | Client <-- Server | Extern <-- Interne

En gros, c'est composant infra qui vas redistribuer les request de **n'import quel** client ver un group de server
Utilser en web infra pour géré plusieur server
Can be use

- Load balance: receive all traffic and distribute the requests on a cluster of several identical Web server instances.
- Cache: can keep in cache responses from servers for a certain amount of time - if the same request is received again, the reverse proxy can return the cached response instead of forwarding the request to the server.
- Encrypt and decrypt traffic: manages secure HTTPS connections with clients and unsecure HTTP connections with servers.
- Protect servers from attacks (e.g. DDoS, SQL injection, etc.): By filtering requests, reverse proxy can protect servers from attacks.
- Serve static content (e.g. images, videos, etc.): can serve static content from a cache.
- Serve multiple domains on the same IP address: can use the Host header to forward requests to the correct server.

Traefik is an open-source Edge Router that makes exposing/publishing your services on the Internet a fun and easy experience.

Load balancing

Distribution of a set of tasks over a set of ressources (computing units)
With Reverse proxy and Host header => load balancer can be used

A load balancer can distribute requests between multiple servers using different strategies:

- Round-robin: the load balancer forwards requests to each server in the pool in turn.
- Sticky sessions: the load balancer forwards requests from the same client to the same server with the help of a cookie.
- Least connections: the load balancer forwards requests to the server with the least number of active connections.
- Least response time: the load balancer forwards requests to the server with the least response time.
- Hashing: the load balancer forwards requests to the server based on a hash of the request (e.g. the IP address of the client, the URL of the request, etc.).

Type of caching

- Client side (private)
- Server side (shared)

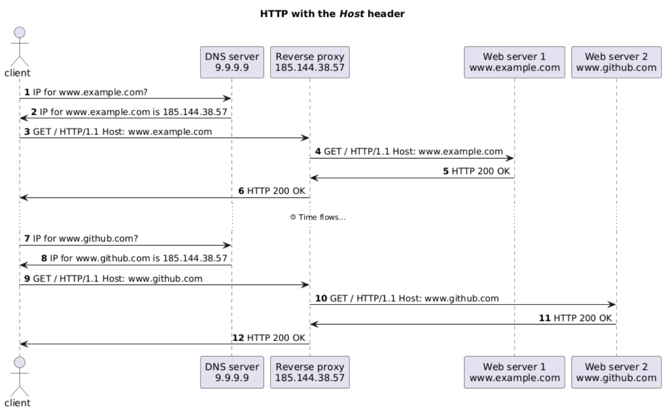
Caching data & Performance

CDN (Content Delivery Networks)

C'est un type de chache que le server peu utiliser pour le contenue static (image, video, etc.)

En gros, c'est un réseau de proxy server qui permette de server les request avec le contenue static recurrent.

- CDN is a geographically distributed network of proxy servers and their data centers.
- CDN can be used to improve the performance of a system by serving static content to clients from the closest server for clients all around the world.



Managin cache with HTTP (server side)

Caching model (on peut utiliser le 2 en meme temps)

RFC 2616

Expiration model

The cache is valid for a certain amount of time.
On peut l'implementer avec le header `Cache-Control: max-age=<number of seconds>`
Validation model

The cache is valid until the data is modified.

Main idea

1. Send a request to the server to check if the data has changed.
2. If the data has not changed, the server can return a `304 Not Modified` response to the client. The client can then use the cached data.
3. If the data has changed, the server can return a `200 OK` response to the client with the new data.

La request s'appel une `conditional request (CR)`
Si un client veu update une ressource il doit envoyer un CR
Reponce pour le **POST** :
`200` = OK = Cache HIT
`412` = Precondition Failed = data all ready modify = Cache MISS

2 type de conditional request (CR)

- Based on the Last-Modified header: 304 si la donner à pas été changer depuis la dérnière fois
- Based on the ETag header: 304 si la hash/version de la donner à pas changer

Header `Last-Modified` (HTTP)

`Last-Modified`: indicates the date and time at which the resource was last updated.

Header `Etag` (HTTP)

`ETag` : provides the current entity tag for the selected representation. Think of it like a version number or a hash for the given resource.

Managin cache with proxie

Forward / Reverse proxy peuvent mange le cache
`Treafik` est http reverse proxy qui peut gerer le cache

Managin cache with key-value stores

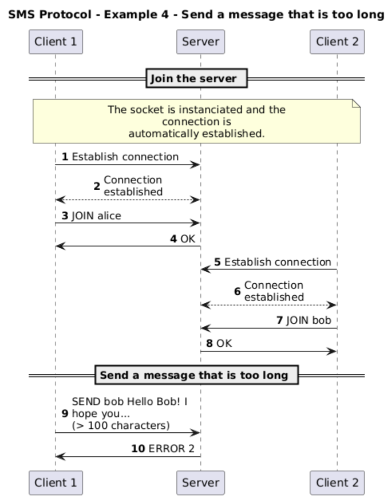
A key-value store is a type of database that stores data as a collection of key value pairs.
Ex : `Redis` or `Valkey`

API design

CRUD = Create - Read - Update - Delete

API description

```
# Title
{Des}
## Endpoints
### {request name}
- {url} | ex : `POST /users`
{desc}
##### Request
{desc}
- {params}
...
##### Respond
{desc}
- {params}
...
#### Status codes
- `201` (Created) - ...
- `400` (Bad Request) - ...
- `409` (Conflict) - ...
```

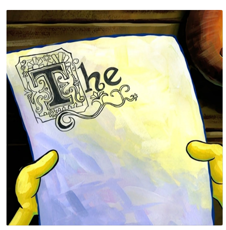


Web Infra

Host Header

header use in **HTTP** fto specifies the domain name of the server (send by client) => making multiple domains in the same IP possible (with reverse proxy helps)
The reverse proxy recieve the client request and send it to the right web server

Avantage du cache
- Reduction
  - Latence
  - Bande passante
  - charge sur les serveurs
  - couts
- Amelioration
  - disponibilité
  - securité



Pour les bytes[]

Example Output (pour écrir sur un fichier) stream (**FOR BYTES[]**) :

```
OutputStream fs = new FileOutputStream("file.data");

BufferedOutputStream bos = new BufferedOutputStream(fs);

for (int i = 0; i < 256; i++) {
  bos.write(i);
}

// flush le buffer -> écrit le buffer dans le fichier
bos.flush();

// close le stream, Close un buffer close aussi le stream
bos.close();
```

Example Input (pour lire sur un fichier) stream (**FOR BYTES[]**) :

```
InputStream fs = new FileInputStream("file.data");

// utiliser un buffer pour tous charger -> lire
BufferedInputStream bis = new BufferedInputStream(fs);

int b;
while ((b = bis.read()) != -1) {
  System.out.print(b);
}

// close le stream, Close un buffer close aussi le stream
bs.close();
```

C'est fonction peuevent retourner une **FileNotFoundException**

Pour les String

Y a une sous type de stream qui permette de géré les charactère facilement

```
Reader reader = new FileReader("file.data", StandardCharsets.UTF_8);
```

et le buffer

```
BufferedReader br = new BufferedReader(reader);
```

Example de lecture de string

```
String line;
while ((line = br.readLine()) != null) {
  // Careful: line does not contain end of line characters
  bw.write(line + END_OF_LINE);
}
```

```
Writer writer = new FileWriter("file.data", StandardCharsets.UTF_8);
```

et le buffer

```
BufferedWriter bw = new BufferedWriter(writer);
```

après ça s'utilise comme le stream de base

BREF, faux just foutre tous ça dans un **try/catch** pour choper les **IOException**

TCP

Client

```
// declare socket, writer, reader
try (Socket socket = new Socket(HOST, PORT);
      Reader reader = new InputStreamReader(socket.getInputStream(), StandardCharsets.UTF_8);
      BufferedReader in = new BufferedReader(reader);
      Writer writer = new OutputStreamWriter(socket.getOutputStream(), StandardCharsets.UTF_8);
      BufferedWriter out = new BufferedWriter(writer); ) {
    // ...
} catch (IOException e) {
  // ...
}
```

Server

```
// declare server socket
try(ServerSocket serverSocket = new ServerSocket(PORT)){
  // ...
}catch(Exception e){
  // ...
}

// declare reader/writer
try (
    Socket socket = serverSocket.accept();
    Reader reader = new InputStreamReader(socket.getInputStream(), StandardCharsets.UTF_8);
    BufferedReader in = new BufferedReader(reader);
    Writer writer = new OutputStreamWriter(socket.getOutputStream(), StandardCharsets.UTF_8);
    BufferedWriter out = new BufferedWriter(writer)) {
      // ...
  }catch (IOException e) {
    // ...
}
```

UDP

Client

```
// declare socket
try (DatagramSocket socket = new DatagramSocket()) {
    // Get the server address
    InetAddress serverAddress = InetAddress.getByName(HOST);

    // Transform the message into a byte array - always specify the encoding
    byte[] buffer = MESSAGE.getBytes(StandardCharsets.UTF_8);

    // Create a packet with the message, the server address and the port
    DatagramPacket packet = new DatagramPacket(buffer, buffer.length, serverAddress, PORT);

    // Send the packet
    socket.send(packet);

    System.out.println("[Client] Request sent: " + MESSAGE);
} catch (Exception e) {
  System.err.println("[Client] An error occurred: " + e.getMessage());
}
```

Server

```
// declare socket
try (DatagramSocket socket = new DatagramSocket(PORT)) {
  while (!socket.isClosed()) {
    // Create a buffer for the incoming request
    byte[] requestBuffer = new byte[1024];

    // Create a packet for the incoming request
    DatagramPacket requestPacket = new DatagramPacket(requestBuffer, requestBuffer.length);

    // Receive the packet - this is a blocking call
    socket.receive(requestPacket);

    // Transform the request into a string
    String request =
        new String(
            requestPacket.getData(),
            requestPacket.getOffset(),
            requestPacket.getLength(),
            StandardCharsets.UTF_8);
  }
} catch (Exception e) {
  // ...
}
```

cast

- Broadcast : Envoie a tous le monde ans le reseau

```
// add this
socket.setBroadcast(true);
// boardcast adress sois un subnet mask
InetAddress serverAddress = InetAddress.getByName("172.25.255.255");
```

- Multicast : Envoie a tous le monde qui correspond au masque
  Sender

```
// Init multi cast address
InetAddress serverAddress = InetAddress.getByName("239.0.0.0");
```

Recever

```
// faux crée un MulticastSocket
try (MulticastSocket socket = new MulticastSocket(PORT)) {
  // Crée le group
  InetSocketAddress multicastGroup = new InetSocketAddress("239.0.0.0", PORT);
  // set un nom
  NetworkInterface networkInterface = NetworkInterface.getByName("NomRandom");
  // join le group au socket
  socket.joinGroup(multicastGroup, networkInterface);
}
```

Thread

```
public Integer SuperThreadFunc() {
  // Manage emitters
}
```

Define "thread" function

```
try (ExecutorService executorService = Executors.newFixedThreadPool(2); ) {
  executorService.submit(this::SuperThreadFunc);
  executorService.submit(new ThreadClass(/*...*/));
  //...
} catch (Exception e) {
  System.out.println("[Receiver] Exception: " + e);

  return 1;
}
```

crée un Executor service

Special type

- y a les `AtomicBoolean` , `AtomicInterger` , ...
- et les `ConcurrentHashMap` , `ConcurrentLinkedQueue`